Exp No: 14

Date: 08/10/25 Write a code using RAW Sockets to implement packet sniffing.

code:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

    protocol_name = ""
    if protocol == 1:
        protocol_name = "ICMP"
    elif protocol == 6:
        protocol_name = "TCP"
    elif protocol == 17:
        protocol_name = "UDP"
    else:
        protocol_name = "Unknown Protocol"
    print(f"Protocol : {protocol_name}")
    print(f"Source IP : {src_ip}")
    print(f"Destination IP: {dst_ip}")
    print("_" * 50)

sniff(iface = "Wi-Fi", prn = packet_callback, filter = "ip",
      store=0).
```

Destination IP: 224.0.0.251

Protocol: TCP

Source IP: 172.16.11.15

Destination IP: 172.16.8.1

Protocol: UDP

Source IP: 172.16.11.136

Destination IP: 239.255.255.250

Result:

Thus, the python was successfully executed using python socket programming